

Designing for Accessibility, Low Literacy, and Low Connectivity

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Abstract

This dissertation investigates whether the Zavala County, Texas e-government website, redesigned for users with accessibility needs, low literacy, low connectivity, or other barriers, produces measurable improvements in usability. The prototype is a static site without a live database or backend, as the goal was to evaluate accessibility rather than to develop a deployable government website. The literature review identifies specific gaps between design guidance and outcome evidence, with e-government research documenting features and deployment decisions over addressing accessibility gaps directly. The prototype prioritized plain language content, simplified navigation, a real-time search function, and a Spanish-language version of each webpage, informed by demographic data showing that over 90% of Zavala County residents identify as Hispanic or Latinx. AI features such as chatbots and voice navigation were excluded, since the literature supporting their effectiveness is largely unevaluated. Five participants evaluated usability by comparing the original and prototype. Task success rate improved, and mean completion time was roughly halved on the prototype. These findings support the design choices made, though the sample size and short-term evaluation window limit how far the results generalize. The study concludes that targeted simplification produces measurable gains for users, presenting a more testable alternative to AI accessibility interventions.

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Special thanks to my advisor for guidance throughout this project, especially during the shift from an evaluation-only design to an iterative, user-involved redesign process. I am also grateful to the five participants who took part in the usability evaluation. Their time, feedback, and willingness to work through both versions of the website made this dissertation possible.

Statement of AI Usage

Artificial Intelligence was used in the preparation of this dissertation to assist with identifying reputable peer-reviewed resources and reorganizing the structure of the report by suggesting additional subsections and relevant areas of discussion. AI was also used to support schema design based on manually recorded task-protocol sessions and to assist in translating the resulting data into PostgreSQL development.

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1. Introduction

1.1. Background and context

Specific populations globally find navigating e-government websites and digital spaces challenging and time consuming. E-government websites should prioritize simplicity in design, avoiding unnecessary or excessive content. The focus should be on delivering only the information that is directly relevant and essential to the services and programs offered. Inaccessible web content means that individuals with disabilities or those who find navigating online spaces are denied equal access to information.

The purpose of this project is to redesign the Zavala County Texas website, a real county level government website, to meet WCAG (Web Content and Accessibility Guidelines) and ADA (Americans with Disabilities Act) standards and to optimize the website for accessibility by incorporating recommended approaches to website design and development from literature and reputable sources. The prototype is not to replicate all the content from the original website, but to demonstrate an effective way to present and organize the necessary information for specific users. The final prototype of this project is to address the needs of first-time visitors and individuals with accessibility challenges, low literacy, and low connectivity.

Following the design of the prototype, user testing was conducted to assist with any following modifications that were implemented based on user feedback, following the recommended best practices for website design as supported by literature. User testing consisted of task-based questions and were analyzed to identify user concerns and support the remediation process of the website.

1.2. Aim and objectives

This project aims to redesign and evaluate an accessible, low-literacy-optimized public e-government service interface to provide evidence-based insights into design techniques and accessibility standards that improve task completion for underserved and digitally excluded user groups. In the proposal, six objectives and work packages were identified to support the research, design, and implementation of the prototype. In short, the following are the objectives discussed in the proposal:

- O1/WP1: Literature review of accessibility, low-literacy, and low-connectivity design guidelines, identifying a minimum of 5 design principles.

- O2/WP2: Prototype design implementing at least 4 measurable accessibility improvements.
- O3/WP3: Database design capturing user testing sessions.
- O4/WP4: Usability testing with 5-10 participants.
- O5/WP5: Data analysis of both qualitative and quantitative metrics and feedback from user testing sessions using visualization and query tools.
- O6/WP6: Evaluation and reflection of design decisions and the prototype's effectiveness and limitations.

Four of the six objectives proposed were fully achieved. Objectives O3 and O5, however, were adjusted to better support the enhancement of the prototype and the project's overall goals, aligning with literature recommendations on user testing for website design.

1.3. Dissertation structure

This report expands on the literature review introduced in the proposal, offering a deeper synthesis of recommendations regarding appropriate e-government website design and a focused discussion of the key themes and insights. The preparation and design section discusses methods used to assess the original Zavala County website and focuses on initial preparation and design of the prototype. The implementation and evaluation section further details the adjustments made to the overall project and the objectives as discussed in the introduction. The section also explains the preparation for user testing, the evaluation methodology, and user testing results. The redesign modification section outlines any changes made to the initial prototype post-user testing and evaluation. The report concludes by summarizing the findings, limitations, and recommendations for future work.

2. Literature Review

2.1. Digital exclusion and the case for e-government reform

The need for accessible e-government design is critical because many people continue to face barriers when using digital public services. Although global literacy rates have improved over time, 739 million adults worldwide remained illiterate in 2024 (UNESCO, 2025). In addition, 40% of the global population does not have access to education in a language they understand, which demonstrates the magnitude of language related exclusion in public information systems (UNESCO, 2016). These figures show that many users may struggle to read, interpret, or act

on online government information, particularly when services are designed using complex language, dense layouts, or unfamiliar administrative verbiage.

This issue is important as many governments have adopted a digital-first approach to service delivery. While e-government can improve efficiency and access, it can also exclude users who have low literacy, low digital confidence, language barriers, disabilities, low connectivity, or difficulty navigating complex interfaces. Research shows that effective e-government experiences depend on usability, content quality, accessibility, privacy, performance, personalisation, and available support channels (Alazemi and Al-Shehab, 2024). However, many government websites continue to present accessibility problems, including poor compliance with Web Content Accessibility Guidelines (WCAG), which can prevent users from completing essential tasks online (Aljedaani, 2025).

The problem is whether government information is understandable, usable, and accessible to the people who need it. Users with low literacy, mild intellectual disabilities, limited reading comprehension, visual sensitivity, or different language needs may experience digital public services as confusing or overwhelming (van Calis et al., 2025). For these users, unclear navigation, excessive text, hidden menus, inaccessible layouts, and untranslated information can create practical barriers to accessing services.

This matters because e-government websites are often used to access essential public information, including health, community, administrative, and civic services. If these websites are not designed with accessibility, language, and usability in mind, they risk excluding the very groups who may rely on them most. Therefore, this redesign needs to focus on simplifying information, improving navigation, supporting multilingual access, and creating a clearer user experience for users who may otherwise be disadvantaged by conventional government website design.

2.2. Simplification and cognitive load reduction

Research on digital inclusion suggests that simplified, user-centered digital layouts and experiences are particularly important for vulnerable populations. Morte-Nadal and Esteban-Navarro identify low digital and administrative literacy, complex procedures, technical jargon, and language accessibility as key barriers to engagement with digital public services, highlighting the need for interfaces that reduce cognitive and linguistic demands on users. These findings are especially relevant to low-literacy adults, migrants, and ESL users, those

who navigate complex digital spaces and may be presented with substantial challenges (Morte-Nadal and Esteban-Navarro, 2025).

Caution is required when applying all design recommendations derived from other vulnerable populations. In particular, the guidelines proposed by Gomez-Hernandez et al. were synthesized primarily from usability studies involving older adults. Their recommendation to simplify interfaces is grounded in evidence that age-related cognitive changes can make complex online spaces more difficult to understand and remember. While the recommendation to increase the size and spacing of interactive controls is primarily motivated by age-related declines in visual sharpness, dexterity, and motor control (Gomez-Hernandez et al., 2024), these recommendations are framed within a Universal Design perspective, which is defined as the design of products, environments, programs, and services to be usable by everyone to the greatest extent possible without the need for adaptation. The underlying evidence of the study is largely drawn from users aged 60 years and older. With nuance and discernment, it's important to recognize that not everyone requires extreme adjustments, such as font sizes significantly larger than the standard recommended size.

The recommendation to simplify interfaces appears transferable to low-literacy adults, migrants, and ESL users because it directly addresses barriers identified in the digital inclusion literature, including limited digital skills, language difficulties, and administrative complexity (Morte-Nadal and Esteban-Navarro, 2025). By contrast, the case for enlarged controls is less clear. The rationale for larger touch targets is primarily associated with sensory and motor limitations rather than literacy or language related challenges (Gomez-Hernandez et al., 2024). For low-literacy adults, migrants, and ESL users, design features such as plain language, multilingual support, step-by-step guidance, visual cues, and reduced procedural complexity may be more directly relevant to their needs. Therefore, while interface simplification can be reasonably justified for a particular target population, extending the same recommendation for enlarged controls to low-literacy adults, migrants, and ESL users involves an assumption that is not directly supported by the available evidence.

Furthermore, Islam et al.'s (2023) systematic review of interfaces for illiterate and semi-literate users identified simplicity as one of the most important design principles, resulting in guidelines covering text use, interaction patterns, pictography, audio-video content, and information architecture. Similarly, Guimaraes et al. (2023) concluded that interfaces for low-literacy users should be simple, clear, and objective, with design decisions explicitly informed by the difficulties experienced by the target population.

Several of the strongest studies derive their recommendations from direct engagement with the target users. Gomez-Hernandez et al. (2024) included only studies that generated guidelines through usability testing with older adults, while van Calis et al. (2025) combined interviews and usability testing with people with low literacy and mild intellectual disabilities to identify inclusive design requirements. Their findings demonstrate the value of observing real user interactions, uncovering barriers that may not be apparent through theoretical guidelines alone.

2.3. Universal and inclusive design

Abdurahman and Kabanda (2024) define inclusive design as the integration of the full range of human diversity into the design process to address accessibility challenges. However, this concept remains broad and lacks clear operational definitions. It does not specify which dimensions of diversity should be prioritized or how designers can determine when these dimensions have been sufficiently addressed. Dosis (2014) provides a more concrete explanation by listing categories such as ability, language, culture, gender, age, and other forms of human difference. Nevertheless, this categorization identifies what constitutes diversity without detailing how design processes should incorporate these aspects in practice.

Calis et al. (2025) offer one of the few concrete operational examples in this literature: webpages that visually represent diversity through illustrations of children, older people, and people with physical disabilities. This suggests one route to apply inclusive design, representation in visual content, but it's a narrow slice of what "full range of human diversity" would need to cover if taken literally.

2.4. Multimodal and multilingual support

The proposal highlights voice navigation and real-time translation as features to assist non-native speakers and users with visual impairments. This focus reflects broader trends in government initiatives. Microsoft's AI for Accessibility program works with government agencies to develop tools such as speech-to-text, real-time captioning, and adaptive interfaces for individuals with visual and motor impairments. Canada's AI assistants help citizens navigate government websites, and the EU has partnered with Microsoft to deploy AI sign language interpretation tools for hearing-impaired users (Djatkiko et al., 2025).

These examples in the proposal described deployment and policy framing, not outcome evaluation. No source cited assesses whether these initiatives improved accessibility or user experience. The claims may rest on productivity metrics rather than measurable gains for users, particularly in disability-related or citizen-facing sign language contexts. This points to a wider

gap in the literature: assertions about AI narrowing accessibility gaps in e-government are often supported by initiative announcements rather than independent evaluation.

Glasser (2025) offers a related critique, arguing that AI accessibility tools are frequently built through a technology-first approach that fails to address the actual needs of the targeted communities. This is especially visible in sign language technology development, a prominent area in ML and AI research. Glasser's proposed alternative is a user-first design process that involves end users throughout development, interface design, and testing, a direct response to the same evaluation gap identified above.

That gap in evaluation carries over into chatbot deployment, where the literature is further along. Papadopoulos et al. (2025) categorize government chatbots into two functional types: informative chatbots, which provide policy information, emergency alerts, real-time updates, and public service guidance; and service-completion chatbots, which handle administrative tasks such as scheduling, form submission, and application processing. Papadopoulos et al. (2025) argue that chatbot deployment may improve accessibility in interactions but introduces technical, legal, and ethical risks requiring legal frameworks, interdisciplinary oversight, and proactive mitigation.

Chen et al. (2023) add to this by examining chatbot adoption across 22 U.S. state governments, distinguishing what drives adoption from what determines successful implementation. Their case study, based on the sampling of state-level agencies, identifies four levels of adoption drivers. At the innovation level, chatbots address common questions and reduce workload, though this raises a question worth pursuing, whether some agencies adopt chatbots primarily to project a tech-forward image rather than to meet a demonstrated need. At the environment level, adoption is shaped by citizens, the IT industry, and other government agencies, with COVID-19 acting as an early shock that accelerated uptake. At the organization level, leadership recognition of chatbot value is a prerequisite for any project moving forward. At the individual level, staff exposure to chatbots is shaped by their own prior experience with the technology. Chen et al. (2023) also identify concrete benefits, including 24-hour service and multilingual support that reduce staff workload and expand access across service lines, citing EMMA, deployed by the U.S. Department of Homeland Security, as an example case.

These benefits come with three risks Chen states directly. Accuracy risk stems from chatbots' design, which can produce inaccurate or fabricated output, commonly called hallucinations (Papadopoulos et al., 2025). For e-government platforms, this is a direct concern. Incorrect

information can mislead users, particularly in high-stress situations. Privacy risk arises because chatbots process sensitive personal data during interactions, which carries heightened exposure in a U.S. context. Security risk follows from the fact that government chatbots are a target for cyberattacks.

Trust in these systems is not uniform. Aoki (2020) found that public trust in government chatbots is contextual. Trust is lower for tasks requiring empathy or situational judgment and higher for procedural tasks. Papadopoulos et al. (2025) extend this into a design standard, arguing that a chatbot's value is tied to its inclusivity and should be evaluated directly against WCAG compliance and multilingual capability. This supports a key point that counters the proposal's framing. A chatbot that does not clearly serve its purpose is not neutral. It introduces friction or user frustration, particularly if it persists on every page or disrupts navigation. Papadopoulos et al. (2025) organize chatbot evaluation into five categories:

Core functionality covers input comprehension, intent recognition, response accuracy, and consistency. **User experience** covers conversational fluency, context awareness, error handling, and response timing. **Ethics and safety** cover bias mitigation, ethical compliance, and data protection and privacy. **Adaptability** covers domain versatility, integration, scalability, and maintainability. **Inclusivity** covers multilingual support, accessibility, personalization, and service automation.

The literature on AI-driven accessibility in e-government is heavier on deployment than on evaluation, which tracks with where the technology is right now. Agencies are still focused on rolling AI out. Chatbot adoption is documented in more depth than multimodal accessibility features, but even there, the evidence favors certain motivations (why agencies adopt) over outcome data (whether users are better served).

2.5. Offline-first and connectivity-resilient architecture

Pothineni (2024) and Josephe et al. (2023) propose progressive web applications (PWAs) as a solution to unreliable connectivity, arguing that cached assets let a site remain usable or partially usable when a network drops. This is because cached assets, such as HTML, CSS, JavaScript, and images stored locally in a user's browser, allow the browser to load these assets from its cache instead of requiring a live internet connection. The study's participant pool draws from general urban users with stable broadband access for baseline comparison. This approach was not observed with rural users on inconsistent mobile data, with variable digital literacy. A

PWA's offline caching behavior degrading for a user who has never encountered a "you are offline" message before remains untested in this literature.

A significant challenge in development communication is that the audiences with the greatest need for online information often have access only to low-bandwidth internet connections. Despite this reality, many continue to design websites that assume reliable, high-speed connectivity, limiting accessibility for users in resource-constrained environments. To address this issue, website development should adopt a user-centered approach that prioritizes the needs, behaviors, and circumstances of the intended audience. This involves consulting users to identify the information they value most, understanding demographic characteristics such as age, occupation, technical skills, and interests, and involving representative users in testing website functionality under actual low-bandwidth conditions. Effective low-bandwidth design should enable users to locate essential information quickly through clear site architecture and intuitive navigation, reducing the need to load unnecessary pages. Users should be able to understand a page's purpose and navigate it within approximately fifteen seconds, with key information and navigation elements positioned at the top of the page so they are visible before the page fully loads. Technical optimization is equally important. Web pages should remain lightweight to ensure acceptable loading speeds on slow connections. Developers should minimize the use of images, optimize any images that are required, and ensure websites remain functional even when images are disabled. Cascading Style Sheets (CSS) should be used efficiently in place of image-based layouts, while excessive JavaScript, embedded styles, and table-based layouts should be avoided. Additional performance improvements can be achieved by reducing HTTP requests, enabling server-side compression, supporting browser caching, and limiting the use of PDF documents unless they have been specifically optimized for low-bandwidth environments. Where large downloads are unavoidable, file sizes should be clearly displayed so users can make informed decisions about data usage. Finally, websites should prominently display publication dates, deadlines, and project timelines to help users assess the relevance of information quickly. Collectively, these measures improve accessibility, usability, and information equity for users operating under limited bandwidth conditions (Colmery, 2008).

2.6. Accessibility standards

In terms of accessibility compliance frameworks, WCAG 2.1 Level AA is the technical standard (Department of Justice's 2024 mandate for all government websites). It gives organizations clear technical benchmarks, but meeting them does not guarantee that a site is

usable (ADA.gov, 2024). Local governments have moved toward ADA-compliant practices, though consistent developer training and ongoing maintenance remain persistent obstacles.

Aljedaani (2025) evaluated 64 U.S. e-government websites against WCAG 2.2 automated testing criteria, checking both minimum conformance (Level A) and recommended conformance (Level AA). WCAG defines three conformance tiers. Level A is the baseline: it fixes the most severe barriers, such as missing alt text or content that traps keyboard users, but a Level A site can still be hard to use for many people with disabilities. Level AA is the level most laws and policies require, including the ADA (Americans with Disabilities Act), Section 508, and EU accessibility directives. It includes everything in Level A plus minimum color contrast ratios, resizable text that doesn't break layout, consistent navigation, and visible focus indicators. Level AAA is the strictest tier and is rarely mandated in full, since some of its criteria are difficult to satisfy for certain content types, such as sign language interpretation videos.

Aljedaani (2025) found accessibility problems widespread across the sample, cutting across the core principles of perceivable, operable, and understandable content in ways that could obstruct users with disabilities. Sites tested against higher conformance levels sometimes showed more identified errors. This sounds backwards at first, but the explanation is clear. The more criteria tested means more opportunities to fail. Aljedaani interprets the pattern as evidence that meeting a baseline standard does not necessarily reflect deeper accessibility consideration in the design process, and suggests accessibility is often treated as a compliance checkbox rather than a design priority.

A separate review of user experience in e-government services offers practical guidance for practitioners (Aldrees and Gračanin, 2023). It recommends understanding the social and environmental context shaping user decisions, grounding design in direct investigation of user needs rather than assumption, and distinguishing between political trust in government and user trust in the technology itself. The review also stresses that usability-testing questions should be vetted by field experts before use with real participants.

Governments, especially local or county governments, often lack the time, money, and staff to address accessibility fully, on top of the other pressures involved in delivering government services. This is the practical case for keeping e-government sites simple while still meeting ADA and WCAG guidelines. Once more, being tech-forward and innovative has value, but without the staff to maintain a more complex system and keep it in compliance, a simpler site built to established accessibility recommendations is the safer choice.

2.7. Readability

Line height, layout width, and text size all affect how easily users can read a page. Clear structure reduces visual clutter, which matters most for users with accessibility needs but benefits everyone. Specific recommendations from the literature suggest using adequate spacing between lines, choose clean typefaces, avoid all caps, avoid underlined text (except links), align text left, and support text resizing. Bold is appropriate for emphasis; full capitalization is not, as it reduces readability (Guimaraes et al., 2023).

There's ongoing discussion about whether to include specifically designed dyslexia friendly fonts, such as OpenDyslexic or Lexend, in website design, even as an optional feature. The priority should be ensuring the website is structurally accessible for all readers, including those with disabilities. There are various fonts available to use that are friendly to those with reading difficulties that still carry the same appearance of formality as used in government websites. Rather than relying solely on a toggle feature, the website should use typefaces that are already accessible such as sans-serif options (e.g., Arial, Helvetica, Roboto) rather than serif fonts (e.g., Times New Roman, Georgia, Garamond). Even if a font isn't specifically designed for dyslexia, its clarity and simplicity can still make it a strong choice for improving general readability (Wery and Diliberto, 2017).

2.8. Layout and navigation

A single-column layout reduces the amount of visual scanning required, which helps users process content in a clear sequence. Guimaraes et al., (2023) refers to Nielsen's (2005) recommendations on this approach alongside keeping options visible on screen rather than hiding them in submenus.

Research highlights a divide in the use of moving versus static elements in web design. Nielsen (2005) contends that moving elements can confuse users and create unnecessary distraction. On the other hand, Kodagoda et al. (2012) propose that such elements may effectively draw attention when used purposefully.

Additionally, hierarchical menus with multiple levels create problems for users with accessibility needs. A linear menu structure is easier to navigate. Reducing total page length and adding shortcut links also helps. Dropdown menus should stay open on click rather than on hover, since users with fine motor impairments may struggle to hold a cursor over a target. Interface elements should be kept to what is necessary. Crowded pages with small tap targets

cause problems on touch devices. Users accidentally tap the wrong element, miss interactive areas, or cannot hold multiple keys simultaneously (Blink UX, 2024).

2.9. Interactive elements

Large buttons with both icons and text labels are easier to use than only icon controls. Icons should be concrete, consistent, and visually distinct from one another. Color should be used conservatively. Animated banners can distract users and should be avoided (Demiris et al., 2001; Ostlund, 2020). Search functions should account for human error such as misspellings, incorrect terms, and other input mistakes.

2.10. Accessibility overlays

Some websites include an accessibility overlay, a widget, usually in a corner of the screen, that gives users additional options such as changing color contrast, adding visible focus rings, or pausing animations. These tools use front-end JavaScript to control presentation elements. Most are provided by third-party vendors on a per-page subscription basis (Hartman and Gorichanaz, 2025).

Overlays can fill specific gaps such as adding missing page language declarations, improving visible focus indicators, adding ARIA labels to form fields, and inserting skip links. However, their effectiveness is widely disputed. The European Disability Forum and the International Association of Accessibility Professionals have warned against treating overlays as a compliance solution (Makati et al., 2024).

Research by Makati, Tigwell, and Shinohara (2004) found that most participants, drawn largely from the blind and low vision community, reported overlays made accessibility worse rather than better. Blind users primarily rely on screen readers, and overlays frequently interrupted screen reader functionality. Many participants stopped visiting sites with overlays altogether. Egger et al. (2022) conclude that overlays do not replace the need for fundamentally accessible design.

Users who are neurodiverse, deaf, or hard of hearing reported more mixed experiences, with some overlay features providing benefit. But even in those cases, overlays consistently fell short when tested against assistive technologies (Makati et al., 2024). Empirical evidence on disabled users' experiences with overlays remains limited, and the practical gap between what overlays claim to fix and what they actually fix for real users has not been adequately studied. Websites that provide a way for the public to report accessibility problems allow website owners to fix accessibility issues (ADA.gov, 2022).

2.11. Synthesis and research gap

Sample sizes across some studies reviewed are small and mostly limited to automated or self-reported data, not observed user behavior. Additionally, some studies involved a targeted group. Evaluation is consistently the weak link across every theme and shows more description of what was built than measurement of whether it worked.

3. Preparation and Design

3.1. Baseline accessibility audit

A baseline audit of the original website was carried out to assess the accessibility standards already in place and to establish a score using the accessibility checker WAVE. The original Zavala County website scored 4.8 out of 10, below average on the scale, with major issues identified. The WAVE evaluation identified significant internal accessibility issues, including missing alternative text, form labels, accessible button names, and problematic HTML elements. The page also contains numerous color contrast errors. These findings indicate that improvements are needed to the site's underlying HTML, content labeling, and CSS. While these issues are technically fixable, the site's overall visual design and usability would also benefit from a broader redesign.

The redesign process began with creating prototype webpages for the homepage, which served as the foundation for the rest of the site. These prototypes were developed using Figma, following preliminary research into effective webpage designs. The redesign drew inspiration from example e-government websites known for their accessibility and usability:

- National Federation of the Blind (NFB)
- USA.gov
- Norfolk County Council (UK)
- Section508.gov
- Healthy Texas Women

To ensure compliance with accessibility standards, these websites were evaluated using WAVE and other accessibility checkers, such as AChecker, to confirm adherence to ADA and/or WCAG 2.2 guidelines. These websites were selected as benchmarks due to their strong design principles addressing accessibility.

3.2. Design justification and methods

This project aligns with common approaches in the field in one area, while diverging from them in another area. The Zavala County prototype was designed as a single case study, not a large-scale evaluation. The original proposal's goal was to target disabled, low-literacy or multilingual (two distinct groups that should not be conflated), and low-connectivity users directly, rather than testing accessibility features on a general population and assuming the results generalize downward. The prototype prioritizes features with measurable accessibility benefits, such as plain language content, a working search function, and functional Spanish language pages, over AI features whose accessibility improvements are claimed but not proven.

The design and development process was structured using a week-by-week approach, following the initial Gantt chart created for the project. This helped to organize the workload, monitor progress, and ensure that each stage of the redesign was completed within the planned timeframe. To support documentation and version control, a folder system was created for each week of development. These folders were labelled in order, such as Week One, Week Two, and Week Three, and were used to store relevant design files, coding progress, assets or images, notes, and updated versions of the website. By week five, the final version was completed using HTML, CSS, and JavaScript. The redesign also uses Bootstrap for structure (e.g., submission forms); JavaScript was used to enhance functionality and interactivity across the webpages. This approach followed similar methods used during recent coursework, where structured file management and version control were important parts of the development process.

The initial development stage began with the creation of an index.html file, which served as the main homepage for the redesigned Zavala County website. This file formed the foundation of the website structure and was used to begin building the layout, navigation, and main content areas. Several visual assets were retrieved from the original Zavala County website to maintain a connection with the existing county identity. These included the county seal or logo, images, infographics, and service-related materials. The purpose of using these assets was to ensure that the redesigned website still reflected the official identity of Zavala County while improving its usability, accessibility, and visual organization.

In addition to Bootstrap, Canva was also used to design the icons for the website, ensuring a consistent visual style and cohesive appearance across all icons. This approach allowed for creative freedom while maintaining a simple, clean, and sleek design. The icons were optimized for screen reader accessibility, aligning with best practices for inclusive design. Additionally,

using Canva ensured compliance with licensing requirements, as no external icon platforms were used without proper permissions.

The titles of existing links from the original website were also reviewed and used to inform the navigation structure of the redesigned version. This was important because the redesign needed to simulate familiarity to existing users while making the navigation clearer and easier to use. Additional pages were then created to support the wider structure of the website. One of the first additional pages developed was a Spanish-language version of the homepage.

Other pages were also developed as part of the redesign, including a health program page. The inclusion of this page was based on the types of services and information commonly provided through the county website. During the research process, it became clear that a minimalistic and simplified webpage layout would be most appropriate for improving accessibility. This approach reduces visual clutter, supports readability, and makes it easier for users to locate important information. A simple design was particularly important because government websites are often used by a wide range of people, including users with different levels of digital confidence, language needs, and accessibility requirements.

As noted in literature review, moving or rotating elements like carousels on homepages increases cognitive load and distracts users. They also add unnecessary complexity, particularly when considering bandwidth constraints and low connectivity. While moving images or carousels can serve a critical role in conveying information in seconds, such as public health announcements or event flyers for government services, by capturing attention within seconds, non-essential movement should be eliminated to reduce cognitive load and enhance usability and connectivity.

A welcoming message was placed near the top of the homepage to create a more approachable and user-friendly experience. At the same time, the homepage was designed to avoid presenting too much information at once. Instead, the focus was placed on displaying the most useful and frequently accessed links. This decision was made to prevent users from feeling overwhelmed and to help them quickly identify the services or pages they are most likely to need.

Overall, the design method followed a structured process. The project began with file organization and homepage development, followed by the collection of relevant assets, navigation planning, and the creation of additional pages. The design choices were guided by accessibility, simplicity, bilingual access, and the need to improve the overall user experience.

Further explanation of specific design decisions, including layout, color, typography, navigation, and accessibility considerations, is provided in the following sections.

3.3. Navigation

The original site's navigation bar included eight links: Home, Departments, District Courts, Probation Department, Commissioners, County Directory, Contact Us, and About Us. These navigation tabs include duplicated content from the dropdown menus found on other tabs, and users are not clearly informed that the information is the same. District Courts, Probation, and Commissioners are all departments, and all of them already appear as entries within the County Directory. Listing them separately in the main navigation created redundant paths to the same information. While offering multiple pathways can help users retrieve information quickly by providing various entry points, it may also introduce confusion. Users might question the accuracy and consistency of the information across all sources, especially if the details are spread across different pages rather than consolidated in one place. This could lead to the need for double-checking information between sources.

For the redesign, the navigation was reduced to four links at the top: Home, County Directory, Contact Us, and About Us. The County Directory now serves as the single point of access for all department and office information, rather than splitting that information across multiple tabs and webpages. This reduction addresses a specific usability concern. Extra navigation items increase cognitive load, and redundant labels are harder to parse for users with low literacy or limited English proficiency. A visitor scanning nine tabs has to determine which one applies to them; several of the original tabs (Departments, District Courts, Probation, Commissioners) could plausibly all seem relevant to the same query, forcing an unnecessary decision. Consolidating them into one directory removes that ambiguity and shortens the path to the information the user needs.

A real-time search bar was integrated near the navigation menu, allowing users to receive instant results as they type. The search engine dynamically suggests the closest matches, ensuring efficient navigation. For instance, typing "clerk" immediately displays relevant options like "county clerk," while entering a name, such as "Michelle Bonilla," retrieves the associated contact details from the directory.

3.4. Spanish-language accessibility and demographic justification

Research focused on identifying regions with low literacy, connectivity, and accessibility challenges, particularly in rural areas of the United States. The south-central and southeastern

regions were prioritized, where gaps in connectivity and accessibility are most pronounced. Border cities like Zavala County, Texas, were highlighted for their predominantly Hispanic/Latinx populations, which often face these challenges.

A Spanish-language version of the webpage was considered an important design choice. According to U.S. Census Bureau, as of July 1, 2025, 92.7% of Zavala County residents identified as Hispanic or Latinx. In addition, 94.1% of residents identified as White, while only 5.3% identified as White alone, not Hispanic or Latinx, demonstrating the importance of distinguishing ethnicity from race within the county's demographic profile (U.S. Census Bureau, 2025). These statistics indicate that Spanish-language access is highly relevant to the local population and should be considered a key part of inclusive government website design.

The inclusion of a Spanish-language homepage was also supported by Zavala County's geographic and cultural context. As a county located in Texas near the U.S. Mexico border, Spanish is commonly spoken among residents and within the wider community. Therefore, providing accessible Spanish-language content is essential for ensuring that residents can understand, navigate, and engage with county information and public services effectively. This design decision supports inclusivity by helping to reduce language barriers and ensuring that key information is available to users who may be more comfortable reading Spanish than English.

Rather than relying solely on automated browser translation tools, such as Google Translate, the redesign considered the need for a dedicated Spanish-language webpage useful for low-connectivity users. Although browser-based translation features allow users to toggle between other languages, automated translations may not always be contextually reliable. Additionally, having the Spanish language option readily available on the webpages is more accessible and efficient than relying on users to navigate to browser-based language settings. This is particularly important for government websites, where information about services, health programs, forms, and public resources must be communicated clearly and accurately. A properly translated Spanish webpage therefore supports clearer communication and reduces the risk of misunderstanding important information.

This approach recognizes the cultural and linguistic needs of the local community. By making key information available in Spanish, the redesigned website aims to create a more accessible and welcoming user experience for Spanish-speaking residents. It also allows more users to access county resources with confidence and ease. Overall, the decision to include a Spanish-

language version of the homepage was justified by both demographic evidence and accessibility considerations, as it reflects the needs of Zavala County's largely Hispanic or Latinx population and supports more equitable access to public information.

3.5. Accessibility toolbar

Subsection 2.10 of the literature review discussed the use of accessibility overlays on websites. Many of these overlays are provided by third parties and are typically implemented as front-end JavaScript widgets that appear on top of an existing website. This website does include an accessibility overlay option in the bottom-left corner of the screen, implemented using front-end JavaScript. However, research has shown that accessibility overlays do not necessarily provide adequate solutions and, in some cases, can create additional barriers or be less beneficial for users who rely on specific accessibility functions.

For this reason, the website should be designed to be accessible from the outset, rather than relying primarily on an accessibility overlay to compensate for accessibility issues. The Zavala County prototype has been designed to follow accessibility principles (WCAG 2.1 AA) directly within the website itself. For example, the recommended font size has been implemented to be readable without requiring users to activate an accessibility toolbar or overlay. The overlay is still available as an additional option for users who may benefit from its features, but it is not being used as a substitute for an accessible website design. Furthermore, the option to have a dark mode and high contrast mode is available for users to assist with comfortability, improved visibility, accessibility, and enhanced focus.

Links are also presented so they are clearly identifiable. For example, the navigation bar uses underlining to indicate the selected page or link, making it clearer to users where they are within the website. The Spanish language option is also clearly presented as a link. In addition, the accessibility toolbar provides an option to underline all links for users who require stronger visual indicators.

This approach is intended to ensure that accessibility is incorporated into the core design of the website rather than being dependent on an overlay. Some websites may rely heavily on an accessibility overlay while still having small fonts, poor link visibility, complex layouts, or other accessibility issues within the underlying website. This can create a distracting experience and does not necessarily address the needs of users with disabilities. In contrast, the redesign aims to make the website inherently accessible through its layout, typography, navigation, and link presentation, while still providing the overlay as an additional support option.

4. Implementation and Evaluation

4.1. Methodological adjustment

Both the literature and user testing were a guide to the methodology of the prototype development. Therefore, a change was made to the initial project proposal. Rather than having evaluators or user testers assess only the final website, they were engaged earlier to provide feedback that would guide further refinements for the final prototype. This adjustment was implemented midway through the evaluation process, after the first two of the five participants had taken part. Given the significant and actionable insights those two participants provided, it was decided to incorporate their feedback and proceed with further refinements after the user testing.

In other words, this meant participants first evaluated a pre-final version of the redesigned website. Their feedback then guided targeted improvements, and the final version, incorporating those modifications, is evaluated and discussed separately in this report, producing a more rigorous and user-informed refinement process than a single final-version evaluation would have allowed. This decision is supported by research emphasizing the importance of involving the target population or users directly in the design process, since such collaboration helps ensure that accessibility features are implemented effectively and that the website meets the needs of its intended audience.

An additional Week 6 folder was created to incorporate modifications based on feedback from the participants. This folder includes all the original elements from the Week 5 final version of the project, but the website within it is a modified version of the Week 5 site, updated to reflect the feedback received.

4.2. Evaluation methodology

The evaluation involved five participants who assessed both the original and redesigned versions of the website. Participants provided informed consent and completed a mixed-methods evaluation, including interviews, observations, and questionnaires. Task-based questions were used to guide their interaction with both sites.

Pre-task and debrief questions were included to gauge participants' perceptions and feelings about each version. Test scenarios required users to locate specific information, such as contact details, service descriptions, or instructions for using accessibility tools. Feedback was documented in real time as participants progressed through each task. Each task was timed to

measure how long it took participants to complete, or fail to complete, the task, while also observing their navigation strategies on both sites.

For Objective 3 of the proposal, which addressed the pulse screen and user interaction logging, the originally proposed automated logging system using a PostgreSQL database was not implemented. Instead, a manual approach was adopted. Task completion times were recorded using a stopwatch, and a database was still used, but only to store participant observations and the manually logged times, rather than capturing real-time user interactions automatically.

4.3. Ethics application and testing procedure

The evaluation process began with submitting an ethics application to the ethics board for approval before conducting the dissertation project. This component involved preparing several key documents, including:

- Participant information sheet
- Consent forms for interviews and observations
- Anonymous questionnaire
- Data collection topics overview
- Participant debriefing sheet

The user testing took place in person and was completed within an hour per participant. Each participant was given a set of task-based questions, along with pre-task questions and closing debrief questions, after evaluating the site.

4.4. Results

4.4.1. Participant profile

Five participants (P1-P5) evaluated the initial redesigned site. All five had used a government or council site before, most commonly for a UK-based government site rather than a US county site: council tax (P3), vehicle/driving-related services (P4), and immigration/visa applications (P5). P2 had looked up taxes and bin collection. P1 recalled visiting a government site previously but not the specific task, though noted having found what was needed at the time. None had prior experience with Zavala County's site specifically, so all five approached both the original and redesigned site cold.

All five are comfortable reading English. Three are monolingual English readers for this purpose (P3, P4) or English plus one other language (P1: Hindi; P2: Turkish). P5 stands out with five additional languages (Tshivenda, Zulu, Sesotho, Tswana, Afrikaans), making them

the sample's clearest case for a non-native English perspective. None of the five read Spanish, so the Spanish-parity task (Task 5) had no participant who could meaningfully evaluate it.

Three participants (P3, P4, P5) reported consistently fast connections on a typical day. Two (P1, P2) reported variable speed depending on location, with P2 noting home connection is fast specifically. None reported a consistently slow connection.

4.4.2. Cold start impressions

Original site (5 impressions): Four of five participants described the homepage as busy or overloaded. Recurring complaints included too much text (P3), oversized images that push content down (P4), an unclear calendar widget (P3, P5), and a moving burn-ban banner competing for attention (P5). P4 also noted a functional problem on first look, the search bar did not work. P1 was the exception, describing a more neutral first impression.

Redesigned site (4 impressions): All four participants who commented described the layout as calmer and easier to parse. Quick links with icons were called out positively by two participants (P2, P4), and no participant described a single oversized image dominating the page, a direct contrast to the original site complaints. P4 separately noted the Spanish-language option as a first-look positive.

4.4.3. Task protocol

The usability testing followed a task-based protocol with 10 defined tasks, 9 administered. Each task maps to a specific design decision in the prototype. **Task 5**, the Spanish-language task, was skipped in every session. As none of the five participants read Spanish, there was no one in the sample who could evaluate whether the Spanish pages are a true parallel structure. The tasks split roughly into four categories. **Findability (Tasks 1-4)**: department lookup, search under a vague term, name-only search, navigation without search. Four of nine tasks target this, making it the most heavily tested area. **Resilience (Tasks 6-7)**: Recovery from a wrong click. **First impressions (Task 8)**: unprompted, open-ended, not tied to a specific pass/fail outcome. **Accessibility and linked content (Tasks 9-10)**: external resource discovery plus the read-aloud feature, and the accessibility toolbar directly.

4.4.4. Quantitative task outcomes

The dataset covers 66 rows across 5 participants and 9 administered tasks, split by website version: 31 original site attempts and 35 redesigned site attempts (redesigned has more rows because Task 10, the accessibility toolbar, only exists on that version).

Success rate: The success rate table tracks each task attempt to determine whether the participant found what they were looking for. The results are categorized as “yes” (they found it), “no” (they didn’t find it), or “not sure” (they found something but couldn’t confirm it was correct). On the original site, 25 of 31 attempts were successful, about 80.6%. On the redesign, 34 out of 35 attempts succeeded, which is 97.1%. The totals aren’t the same because Task 10 (the accessibility toolbar) only exists on the redesign. This adds attempts that don’t have a counterpart on the original site. Also, a few tasks (Task 5) were skipped, which slightly reduced both totals from the full 45 attempts (5 participants x 9 tasks). The redesign nearly turned every attempt into a success. The single redesigned site failure was Participant 5 on Task 1 (nutrition contact), the same task that was hardest on the original site too, where 3 of the 5 “no” results occurred.

TABLE 1: SUCCESS RATE

Version	Yes	No	Not sure	Success rate
Original	25	5	1	80.6%
Redesigned	34	1	0	97.1%

Hesitation: The hesitation table measures whether participants moved straight to the answer or paused and backtracked. On the original site, participants hesitated on 22 of 31 attempts (71%); on the redesign, that dropped to 16 of 35 (46%). Hesitation dropped by about a third on the redesign, but it’s still the single most common outcome. To note, “faster and more successful” shouldn’t be mistaken for “immediately intuitive”; participants often got where they needed to go, just not always on the first move.

TABLE 2: HESITATION

Version	Knew next step	Hesitated / backtracked	Hesitation rate
Original	9 of 31	22 of 31	71%
Redesigned	19 of 35	16 of 35	46%

Readability: The readability table counts how each task attempt was rated on ease of use (easy, okay, or hard), based on participant feedback during the session, separate from whether they succeeded or how long it took. On the original site, ratings were spread across all three levels: 16 easy, 11 okay, and 4 hard. On the redesign, ratings collapsed almost entirely to easy: 34 easy, 1 okay, and 0 hard. The “hard” ratings on the original cluster around two tasks in particular, the

breast/cervical cancer screening link (Task 9, flagged hard by 3 of 5 participants) and finding the health program page task (Task 6), meaning the original site's readability problems weren't spread evenly but concentrated in a couple of specific pain points that the redesign resolved.

TABLE 3: READABILITY

Version	Easy	Okay	Hard
Original	16	11	4
Redesigned	34	1	0

Would give up: The “would give up” table counts how many task attempts ended with the participant saying they would abandon the task rather than keep trying, a self-reported measure of frustration, separate from whether they succeeded. On the original site, 3 of 31 attempts (9.7%) ended this way, spread across different participants and tasks (P2/Task 6, P5/Task 1, P5/Task 4). On the redesign, just 1 of 35 (2.9%) did, Participant 5 on Task 1, the same task that also produced the redesign's outright failure. The redesign cuts give-up instances by more than two-thirds. The redesigned give-up reinforces that Task 1 remains the redesign's clearest unresolved problem area.

TABLE 4: WOULD GIVE UP

Version	Would give up
Original	3 of 31 (9.7%)
Redesigned	1 of 35 (2.9%)

4.4.5. Closing debrief

The most tedious tasks include the Nutrition Center contact (Task 1) and the county clerk task (Task 2). Three participants named the nutrition lookup specifically, mostly on the original site. Two participants also flagged the breast/cervical cancer screening task on the redesigned site. The original site draws the most consistent criticism. The most confusing elements were cluttered homepage, poorly organized quick links, a disliked border, and confusing navigation come up across three participants. The redesign's complaints were narrower and more specific. One participant found page content hidden until scrolling due to the fade-in effect overstimulating and preferred a more static page. Low contrast and dark mode toggle features also needed improvement, particularly in terms of visibility for text and icons. The homepage accessibility message was initially mistaken for an advert.

In comparison to prior government sites used by the participants, four of five participants called the redesign easier to use, citing less information overload, dark mode reducing eye strain, and icons reducing reliance on text. Participant 1 was neutral rather than negative; no one rated the redesign worse. One participant (P5) had no fix to suggest. No single dominant complaint emerged, but suggestions cluster into three areas:

Visual/layout: P1 wanted larger text and a smaller header and homepage image; P3 wanted an improved high contrast and dark mode.

Motion/interactivity: P2 wanted a more static site with fewer effects.

Information architecture: P3 suggested grouping quick links into a folder menu; P4 suggested moving county clerk info under topics and services.

4.4.6. Time analysis

A set of queries was run against the timed data to better understand the usability testing results. Not every query is discussed here, only those most relevant to the evaluation. Table 5 shows the per-task means, which are a more reliable comparison than a single average, since they isolate each task rather than mixing fast lookups with slow searches. Task 10 (accessibility toolbar) only exists on the redesign, mean 6.16s, n = 4, so there is no original site baseline to compare against.

TABLE 5: MEAN COMPLETION TIME PER TASK (ORIGINAL VS. REDESIGNED)

Task	Original mean	Redesigned mean	Difference
1. Ambiguous department lookup	97.70s	55.19s	-43%
3. Name-only search	26.65s	17.11s	-36%
4. Navigate without using search	20.78s	13.25s	-36%
7. Recovery from a wrong click	81.74s	1.44s	-98%
9. Link to other resource	63.07s	11.97s	-81%

Task 7 shows the largest gap, but n_{timed} = 2 for both versions, so it reflects two participants, not five. The original site n = 2 mean (81.74s) is also being pulled up by one slow outlier (160.97s vs. 2.50s for the other), so “98% faster” is a fragile headline number even though the direction is consistent with everything else in the table. Table 6 shows that one exception is Task 2 (finding what a county clerk does), which ran 54% slower on the redesign.

TABLE 6: TASK 2 COMPLETION TIME (ORIGINAL VS. REDESIGNED)

	Original	Redesigned
Mean	41.06s	63.37s
n_timed	4	5

This lines up with the qualitative assessment that multiple participants on the redesign expected the clerk's role description under Topics and Services or the County Directory and had to detour to the About Us tab instead. The information architecture sent them down a longer path even though the underlying search worked fine.

The redesigned tasks took about half the time to complete compared to the original site tasks when looking at the overall mean time across all tasks (Table 7), with averages of 25.25 seconds vs. 50.61 seconds. The difference is based on all timed entries in the dataset. The n_timed counts aren't equal (34 vs. 30) because Task 10 only exists on the redesign; this also lets one slow original site outlier (like the 170.59s Task 1 attempt) pull the whole original mean upward.

TABLE 7: OVERALL MEAN TIME ACROSS ALL TASKS

Version	n_timed	Mean seconds
Redesigned	34	25.25
Original	30	50.61

Table 8 shows the per-participant time saved (original task time minus redesigned task time). Among the 29 paired observations (same participant completing the same task on both versions), 21 of 29 (72%) show the redesign faster; 8 of 29 (28%) show it slower. Every participant is net positive overall.

TABLE 8: TIME SAVED PER PARTICIPANT (ORIGINAL MINUS REDESIGNED)

Participant	Net seconds saved	Tasks faster	Tasks slower
1	+363.62	3	2
2	+60.01	5	2
3	+101.30	5	1
4	+73.30	4	1
5	+103.44	4	2

All five participants come out ahead on the redesign in total time, even though every single one hit at least one task where the redesign was slower. The improvement is consistent, but not uniform across every task for any one person. Task 1 and Task 4 are the most inconsistent, faster for some participants, slower for others, despite Task 1's aggregate mean favoring the redesign overall. The aggregate hides real variation. Participants 2 and 5 were both slower on Task 1 redesigned (P5 got lost on the Texas AgriLife page, a different website; P2 took multiple routes to find that information).

Task 2 is a standout problem. The task is negative for 3 of 4 participants who attempted it on both versions (P2: -26.81s, P4: -82.93s, P5: -11.58s), only marginally positive for P3 (+0.27s). This confirms the task-level finding at the individual level. Task 2 is a genuine, repeated regression and not noise from a single outlier.

Largest magnitude of change (positive and negative):

- Biggest gain: P1/Task 7, +159.42s
- Biggest loss: P4/Task 2, -82.93s

5. Redesigned Modifications

After reviewing participant feedback, a few adjustments were made to the initial prototype. The feedback focused on specific elements rather than the overall website or layout. First, improvements were made to the accessibility toggle feature to clearly show which accessibility options are enabled or disabled. Initially, users couldn't tell which features were turned on. One participant suggested it would be helpful to see which accessibility features are active or inactive when the toggle window is open; and a visual indicator (e.g., highlighting) to show an accessibility feature is enabled when the accessibility window is closed. Now, when the user opens the accessibility toggle, each feature's status (on/off) is displayed. When you close the window, the accessibility button in the bottom-left corner of the page remains highlighted, ensuring users can always see that an active accessibility feature is on without obstructing text or images.

An additional change was made to the page layout on the JavaScript-based end. In the initial prototype, the main content, primarily the text on the page, would fade as users scrolled down after the header banner image. One participant mentioned they preferred a static page layout to reduce confusion. They noted that having no visible text when first landing on the page (until scrolling) could feel unintuitive, even though users would eventually scroll to see the content.

However, they emphasized that the content should be immediately visible rather than requiring users to scroll to find it. To address this, the `apps.js` file was simply modified so that content now remains static, with no fading effects. This ensures the content is laid out clearly from the start, reducing confusion and improving user experience.

Another improvement was made to help users find information about specific departments more easily. For instance, some participants had difficulty locating details about what a county clerk does, as the description wasn't intuitive to find. Many initially checked the county directory, which only provides contact details and department names without role explanations. To address this, direct links were added from the About Us section to the county directory. Now, when users visit About Us to learn about a department, they can also access the county directory and see the relevant contact information automatically. This ensures users can find both the role description and contact details in one place. Given that the top navigation panel has five tabs, it remains intuitive to find the county clerk's description. However, it's best practice to provide multiple pathways to the same information.

6. Conclusion

6.1. Summary of findings

The Zavala County prototype outperformed the original across each measure tested. All five participants finished with a net time saving overall. The results were not uniformly positive. Task 2 (finding what a county clerk does) ran slower on the prototype. Participant 5 also failed Task 1 on both site versions. These findings support the redesign as a clear overall improvement while identifying two specific, addressable areas, Task 2's navigation path and Task 1's findability.

6.2. Limitations

Several key limitations are worth noting in this report including the target audience for user testing, the Spanish-language evaluation, and low connectivity evaluation component. For the initial prototype evaluation, the proposal specifically targeted individuals with accessibility needs and those who may be English as a Second Language (ESL) speakers. To clarify, ESL status is not equivalent to low literacy. However, this group was considered the closest available proxy to assessing how individuals with varying levels of English proficiency might perceive the clarity of information presented on an English-language website and navigate its content.

The small sample size also limited how fully the prototype's functionality could be assessed for users with accessibility needs. Participants were not asked whether they had specific accessibility needs before completing the tasks. This followed from a design decision that accessible design should work for any user, not only users who identify as needing it. If a site requires knowing who has a disability before it becomes usable, the design has already failed the goal of universal access. Testing it as a general use site, rather than screening for a specific accommodation, is a more direct test of whether the prototype is accessible without requiring anyone to disclose a need first. This choice comes with a real limit, and research does support testing directly with the users a design targets. Without knowing participants' accessibility needs, the results can't confirm how the redesign performs for someone who relies on the Spanish-language pages, the accessibility toolbar, or even a screen reader compatibility to complete a task. All five participants used both sites as general users, not as users testing a specific accommodation.

The Spanish version of the site was not evaluated, as none of the participants were Spanish speakers. Additionally, the prototype was not designed as a mobile application and was not tested under degraded bandwidth or low connectivity conditions. However, as described in subsection 2.5 of the literature review, its structure and components were similarly designed to support smooth page loading and navigation, even though this capability has not been practically tested.

Another limitation or challenge involved determining what information should be included or excluded in the prototype from the original county website and what information should be presented to users. As noted previously, the website contained a substantial amount of information, including PDF documents related to county commissioners' meeting minutes, outdated disaster declarations, widgets, and other materials that may not be relevant to users, particularly first-time visitors. Therefore, it was important to prioritize information that users would be most likely to need when visiting a county website. This included contact information for all county departments, the latest news and updates, and basic information about how the county courthouse operates. Providing this information was intended to help users find what they need when visiting the website.

Designing the task protocol was another challenge. As noted in subsection 2.6 of the literature review, it is recommended that usability-testing questions be reviewed by field experts beforehand. The questions used in this study were informed in part by existing research and designed to assess whether the website enabled users to find the information they needed

efficiently and comfortably. However, the questions were not vetted by experts in the field. The questions primarily measured participants' confidence rather than their competence. A participant can be confident navigating the prototype and still take four detours to find, for example, a contact number. Session length was not fixed, meaning sessions varied in duration depending on time on each task and documenting feedback. Participants were also not asked upfront whether they had any accessibility needs. Despite that, a few participants brought up the accessibility toolbar unprompted, specifically the dark mode and font-size options, and described them as useful to have on a website without needing to adjust settings themselves through the browser. That feedback was incidental rather than something the task protocol captured.

6.3. Recommendations for future work

Future work should prioritize testing the gaps this evaluation could not close. A larger and more linguistically diverse sample is the clearest need. Incorporating more individuals with diverse accessibility needs would strengthen the evaluation and provide a more comprehensive understanding of how different users interact with the prototype. Similarly, no participant reported being evaluated under a slow or degraded connection, so the offline-first design decisions were never evaluated under real-world conditions.

6.4. Final reflections

The purpose of the prototype was not to replicate all the information contained on the original website, but rather to demonstrate how the website's layout, structure, and visual presentation could be improved based on established accessibility and web design guidelines. This was especially important given the formal nature of government websites at the state and local levels, where clear organization, accessibility, and ease of navigation are essential for effectively serving a diverse user population. With modifications, all objectives detailed in the proposal were modestly achieved. The literature review identified a consistent gap that e-government accessibility research documents deployment and design guidelines, but rarely measures whether the resulting changes can help the people they target. This project addressed that gap, at a small scale, by building a prototype and testing it against the original website with five participants, as well as making modifications based on their feedback.

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Appendix A: Usability Test Protocol

This is not a walkthrough of the websites. Tasks are built to expose weak points in the redesign.

1. Before starting (moderator notes)

Run each task once per participant. Do not correct a wrong path mid-task. If a participant fails a task, do not skip to the next one immediately. Ask what they expected to happen.

2. Pre-task questions

1. Have you visited a county or city government website before? What were you trying to do?
2. Do you read English, Spanish, or both comfortably? What languages are you proficient in?
3. On a typical day, is your internet connection fast, slow, or does it depend where you are?

3. Task scenarios

Each task is a real-world goal, not a click-path, so the participant has to figure out the route.

Task 1: Ambiguous department lookup

“You would like information regarding the Nutrition Center and the services they provide. Find the contact for the office that handles that.”

Stress point: tests whether users can find where courthouse related information lives.

Task 2: Search under a wrong or partial term

“Find out what a county clerk does and the contact information for the person in charge of county records.”

Stress point: tests real-world scenario search.

Task 3: Name-only search

“Someone told you to contact Michelle Bonilla but didn’t say what department she’s in. Find her contact details.”

Stress point: confirms the name-based search.

Task 4: Navigate without using search

“Without using the search bar, find the page that lists all county departments.”

Stress point: isolates whether the navigation is usable on its own.

Task 5: Spanish-language (Not Evaluated)

(For Spanish-reading participants) “Switch the site to Spanish and find the same court office information from Task 1.”

Stress point: tests whether the Spanish pages are a parallel to the English version.

Task 6: Health program

“Find the health program page and tell me what it offers.”

Stress point: tests real-world scenario search.

Task 7: Recovery from a wrong click

“From the health program page. Get back to where you can start over without using browser back button.”

Stress point: tests whether navigation is forgiving.

Task 8: Cold start, no prompt

“Here’s the homepage. What are your first impressions or what sticks out?”

Stress point: tests first impressions and whether quick links actually orient a new user.

Take 9: Link to other resource

“Find the link that will redirect you to information regarding breast and cervical cancer screening, read the name of the website aloud.”

Stress point: tests whether navigation to new resources and the path to get there is intuitive.

Take 10: Use of an Accessibility Function (Prototype Only)

“Find the feature that underlines all links on the page.”

Stress point: tests whether navigation of accessibility toolbar is easy to apply to the website.

4. Post-task probing questions (ask after each task, not just at the end)

1. Did you find what you were looking for? (yes / no / not sure)
2. Was there a point where you didn’t know what to do next? Where?
3. Was the text on this page easy to read? (easy / okay / hard)
4. If this had been your real situation, would you have given up before finishing?

5. Closing debrief

1. Which task was more tedious, and why?
2. Was there anything on the site that felt confusing, cluttered, or hard to trust?
3. Compared to a government website/s you’ve used before, was this easier or harder to use?
4. If you had one thing to fix on the prototype, what would it be?

Appendix B: Zavala County Prototype Setup Guide

Prerequisites

This is a static site: HTML, CSS, and JavaScript only. No database, no backend server, and no npm packages are required to view the pages.

You will need one of the following already installed:

- Python 3
- Node.js
- Or the Live Server extension in VS Code

Step 1 - Get the Project Files

Clone or download the repository:

git clone <https://github.com/kkennebrew/zavala>

Or, if downloading as a ZIP from GitHub, extract it to a folder.

Open a terminal and navigate to the extracted folder:

cd path/to/repo-folder-name

Step 2 - Start a Local Server

Pick whichever option matches what you installed in prerequisites.

Option A: Python

python3 -m http.server 8000

Option B: Node.js

npm http-server -p 8000

Option C: VS Code Live Server

Right-click index.html in the file explorer and select “Open with Live Server.”

Step 3 - Open in Browser

If you used Option A or B above, go to:

<http://localhost:8000>

If you used Live Server, it will open the correct address automatically.